

Managing Variance: Heisenberg-Weyl Inequality

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Motivation

Some of the main ideas in elementary harmonic analysis are related to understanding the duality between a function and its fourier transform. The Heisenberg-Weyl Inequality is one such duality relating the spread of a function in $L^2(\mathbb{R})$ with the spread of its fourier transform.

Understanding $L^2(\mathbb{R})$

Definition

$L^2(\mathbb{R})$ is the space of functions $f : \mathbb{R} \rightarrow \mathbb{C}$ such that

$$\|f\|_{L^2} := \sqrt{\int_{-\infty}^{\infty} |f(x)|^2 dx} < \infty$$

Understanding $L^2(\mathbb{R})$

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$L^2(\mathbb{R})$ also has a natural inner product defined as follows

$\forall f, g \in L^2(\mathbb{R})$,

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx$$

Note $\|f\|_{L^2}^2 = \langle f, f \rangle$ and $L^2(\mathbb{R})$ is a Hilbert space with respect to this norm and inner product.

The Fourier Transform

Definition

Let $e_n(x) := e^{2\pi i n x}$. $\forall f \in L^2(\mathbb{R})$, let $\hat{f} : \mathbb{R} \rightarrow \mathbb{C}$ where $\forall n \in \mathbb{R}$,

$$\hat{f}(n) = \langle f, e_n \rangle = \int_{-\infty}^{\infty} f(x) e^{-2\pi i n x} dx$$

and

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(n) e^{2\pi i x n} dn.$$

Let $\mathcal{F}$ be the Fourier Transform which takes $f \rightarrow \hat{f}$.

Inverse Fourier Transform

Definition

$\forall f \in L^2(\mathbb{R})$, there also exists a inverse transform $\mathcal{F}^{-1}$ which takes $\hat{f} \rightarrow f$

$$\mathcal{F}^{-1}[\hat{f}](x) = \langle \hat{f}, e_{-x} \rangle = \int_{-\infty}^{\infty} \hat{f}(n) e^{2\pi i x n} dn$$

$C_{\downarrow}^{\infty}(\mathbb{R})$ as a subset of $L^2(\mathbb{R})$

Remark

$C_{\downarrow}^{\infty}(\mathbb{R})$ is the set of infinitely differentiable rapidly decreasing functions on $\mathbb{R}$. Infinitely differentiable means that $f \in C^{\infty}(\mathbb{R})$ and "rapidly decreasing" applies to $f, f', f'', \dots$, and means that

$$x^p \frac{d^q f}{dx^q} \rightarrow 0 \text{ as } |x| \rightarrow \infty$$

for every nonnegative integral p and q .

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Important Fact!

$C_{\downarrow}^{\infty}(\mathbb{R})$ is a dense subset of $L^2(\mathbb{R})$. Most proofs regarding properties of $L^2(\mathbb{R})$ use this fact.

The Fourier Transform is a isomorphism of $L^2(\mathbb{R})$

Theorem

$\mathcal{F}$ is a isomorphism of $L^2(\mathbb{R})$ and $\mathcal{F}^{-1}$ is actually its inverse function.

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Corollary: Plancherel Identity

$\forall f \in L^2(\mathbb{R}),$

$$\|f\|_{L^2} = \|\hat{f}\|_{L^2}$$

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Corollary: Parseval's Identity

$\forall f, g \in L^2(\mathbb{R}),$

$$\langle f, g \rangle = \langle \hat{f}, \hat{g} \rangle$$

Major Identities (Cont.)

Theorem: Effect of Differentiation

$$\forall f \in L^2(\mathbb{R}), \hat{f}'(n) = 2\pi i n \hat{f}(x).$$

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Theorem: Effect of Differentiation

$$\forall f \in L^2(\mathbb{R}), \hat{f}'(n) = 2\pi i n \hat{f}(x).$$

Theorem: Effect of Translation

$$\forall f \in L^2(\mathbb{R}) \text{ and } \forall h \in \mathbb{R}, \text{ let } f_h(x) = f(x - h).$$

$$\hat{f}_h(n) = e^{-2\pi i n h} \hat{f}(n)$$

.

Major Identities (Cont.)

Theorem: Effect of Differentiation

$$\forall f \in L^2(\mathbb{R}), \hat{f}'(n) = 2\pi i n \hat{f}(n).$$

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$$\forall f \in L^2(\mathbb{R}) \text{ and } \forall h \in \mathbb{R}, \text{ let } f_h(x) = f(x - h).$$

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Theorem: Effect of Scaling

$$\forall f \in L^2(\mathbb{R}) \text{ and } \forall h \in \mathbb{R}, f_h(x) = hf(x),$$

$$\hat{f}_h(n) = \hat{f}(n/h)$$

Measure of Variance

Definition

$\forall f \in L^2(\mathbb{R})$, let

$$t_0 = \mathbb{E}_f[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

Define $\sigma^2(f) := \|(x - t_0)f\|_{L^2}^2$ be the variance of a function.

The Heisenberg-Weyl Inequality

Theorem: The Heisenberg-Weyl Inequality

$\forall f \in L^2(\mathbb{R}),$

$$\sigma^2(f)\sigma^2(\hat{f}) \geq \frac{\|f\|_{L^2}^4}{16\pi^2}$$

,

with equality **if and only if** f is a constant multiple of the gaussian, e^{-kx^2} for $k > 0$.

The Proof

For all $f \in L^2(\mathbb{R})$, let $t_0 = \mathbb{E}_f[X]$. Let $F(x) := f(x + t_0)$. Thus $\mathbb{E}_F[X] = 0$. So

$$\sigma^2(f) = \int_{-\infty}^{\infty} (x - t_0)^2 |f(x)|^2 dx = \int_{-\infty}^{\infty} x^2 |f(x + t_0)|^2 dx = \sigma^2(F).$$

Thus without a loss of generality, we can assume the center of mass of f is 0 for the rest of the proof.

The Proof

Because $\sigma^2(f) = \|xf\|_{L^2}^2 > 0$, we may as well suppose

$\sigma^2(\hat{f}) = \|n\hat{f}\|_{L^2}^2 < \infty$ because otherwise there is nothing to prove.

Thus

$$\begin{aligned} n\hat{f} \in L^2(\mathbb{R}) &\implies 2\pi i n\hat{f} = \hat{f}' \in L^2(\mathbb{R}) && \text{(Effect of Differentiation)} \\ &\implies \mathcal{F}^{-1}[\hat{f}'] = f' \in L^2(\mathbb{R}) \end{aligned}$$

The Proof

$$\begin{aligned}4\pi^2\sigma^2(f)\sigma^2(\hat{f}) &= 4\pi^2 \int_{-\infty}^{\infty} |xf(x)|^2 dx \int_{-\infty}^{\infty} |n\hat{f}(n)|^2 dn \\&= \int_{-\infty}^{\infty} |xf(x)|^2 dx \int_{-\infty}^{\infty} |2\pi in\hat{f}(n)|^2 dn \\&= \int_{-\infty}^{\infty} |xf(x)|^2 dx \int_{-\infty}^{\infty} |\hat{f}'(n)|^2 dn.\end{aligned}$$

The last equality is true by the **Effect of Differentiation**.

The Proof

$$\begin{aligned}
 4\pi^2 \sigma^2(f) \sigma^2(\hat{f}) &= \int_{-\infty}^{\infty} |xf(x)|^2 dx \int_{-\infty}^{\infty} |\hat{f}'(n)|^2 dn \\
 &= \int_{-\infty}^{\infty} |xf(x)|^2 dx \left\| \hat{f}' \right\|_{L^2}^2 \\
 &= \int_{-\infty}^{\infty} |xf(x)|^2 dx \left\| f' \right\|_{L^2}^2 \text{ (Plancherel Identity)} \\
 &= \langle xf, xf \rangle \langle f', f' \rangle = \langle |xf|, |xf| \rangle \langle |f'|, |f'| \rangle \\
 &= \langle |xf|, |xf| \rangle \langle \overline{|f'|}, \overline{|f'|} \rangle \\
 &\geq |\langle |f'|, |xf| \rangle|^2 \text{ (Cauchy Schwartz Inequality)} \\
 &= \langle |f'|, |xf| \rangle^2
 \end{aligned}$$

The Proof

$$\begin{aligned}
 4\pi^2 \sigma^2(f) \sigma^2(\hat{f}) &\geq \langle |f'|, |xf| \rangle^2 \\
 &= \left[\int_{-\infty}^{\infty} |xf'(x) \overline{f(x)}| dx \right]^2 \\
 &\geq \left[\int_{-\infty}^{\infty} \Re[xf'(x) \overline{f(x)}] dx \right]^2 \\
 &= \left[\int_{-\infty}^{\infty} \frac{1}{2} (xf'(x) \overline{f(x)} + \overline{xf'(x) \overline{f(x)}}) dx \right]^2 \\
 &= \frac{1}{4} \left[\int_{-\infty}^{\infty} x(f'(x) \overline{f(x)} + \overline{f'(x)} f(x)) dx \right]^2
 \end{aligned}$$

The Proof

We are now stuck at this integral, to get a value for the integral create a sequence $f_n \in C_{\downarrow}^{\infty}(\mathbb{R})$ such that $f_n \rightarrow f$ and

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} (1 + 4\pi^2 n^2) \left| \hat{f}_n(n) - \hat{f}(n) \right|^2 dn = 0.$$

This condition on convergence leads to the fact that f_n converges to f pointwise modulo some technical flourishes.

The Proof

Lemma

$$\forall f \in C^1(\mathbb{R}), \quad \frac{d}{dx} |f(x)|^2 = \frac{d}{dx} (f(x) \overline{f(x)}) = f'(x) \overline{f(x)} + f(x) \overline{f'(x)}$$

Lemma

$\forall f \in L^2(\mathbb{R}), |f(x)| \in L^2(\mathbb{R})$ because

$$\| |f| \|_{L^2}^2 = \int_{-\infty}^{\infty} \|f(x)\|^2 dx = \int_{-\infty}^{\infty} |f(x)|^2 dx = \|f\|_{L^2}^2$$

The Proof

$$\begin{aligned}
 & \int_{-\infty}^{\infty} x(f'(x)\overline{f(x)} + \overline{f'(x)}f(x))dx \\
 &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \int_{-m}^m x(f'_n(x)\overline{f_n(x)} + \overline{f'_n(x)}f_n(x))dx \\
 &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \int_{-m}^m x \frac{d}{dx} |f_n(x)|^2 dx \\
 &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \left([x |f_n(x)|^2]_m^m - \int_{-m}^m |f_n(x)|^2 dx \right)
 \end{aligned}$$

The Proof

$$\begin{aligned}
 & \int_{-\infty}^{\infty} x(f'(x)\overline{f(x)} + \overline{f'(x)}f(x))dx \\
 &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \left([x|f_n(x)|^2]_m^m - \int_{-m}^m |f_n(x)|^2 dx \right) \\
 &= \lim_{m \rightarrow \infty} (m(|f(m)| + |f(-m)|)) - \|f\|_{L^2}^2 = -\|f\|_{L^2}^2
 \end{aligned}$$

$\lim_{m \rightarrow \infty} (m(|f(m)| + |f(-m)|)) = 0$ because otherwise $|f(x)|^2 + |f(-x)|^2$ would not be in $L^2(\mathbb{R})$.

The Proof

Thus,

$$\begin{aligned} 4\pi^2 \sigma^2(f) \sigma^2(\hat{f}) &\geq \frac{1}{4} \left[\int_{-\infty}^{\infty} x(f'(x) \overline{f(x)} + \overline{f'(x)} f(x)) dx \right]^2 \\ &= \frac{1}{4} \|f\|_{L^2}^4. \end{aligned}$$

Finally,

$$\sigma^2(f) \sigma^2(\hat{f}) \geq \frac{1}{16\pi^2} \|f\|_{L^2}^4$$

The Proof

To get the lower bound, we look all the way back to the key application of the Cauchy Schwartz Inequality where $\langle xf, xf \rangle \langle f', f' \rangle \geq |\langle f', xf \rangle|^2$ with equality if and only if $f' = kxf$ almost everywhere where $k \in \mathbb{C}$.

The Proof

The second inequality was the fact that $|xf'\bar{f}| \geq \Re(xf'\bar{f})$. Lets use this condition to deduce a bit more about k .

$$|kx^2|f| = |kx^2f\bar{f}| = |xf'\bar{f}| \geq \Re(xf'\bar{f}) = \Re(kx^2f\bar{f}) = \Re(kx^2|f|)$$

This means $|kx^2|f| \geq \Re(kx^2|f|)$. Equality is only possible if $kx^2|f| \in \mathbb{R}$ and $kx^2|f| \geq 0$ for all $x \in \mathbb{R}$. But since x^2 and $|f|$ are in $\mathbb{R}$ and are greater than or equal to 0. That means $k \in \mathbb{R}$ and $k \geq 0$. But $k \neq 0$, because otherwise f is a constant function and constant functions are not in $L^2(\mathbb{R})$. Thus f is a solution differential equation $f' = kxf$ for $k > 0$. Thus f is a multiple of the gaussian, e^{-kx^2} for $k > 0$.

The Heisenberg-Weyl Inequality

Theorem: The Heisenberg-Weyl Inequality

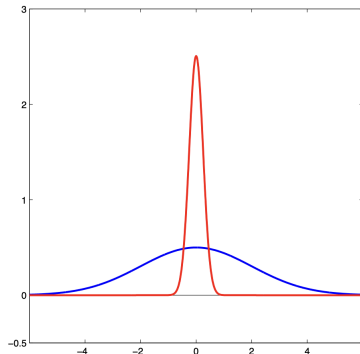
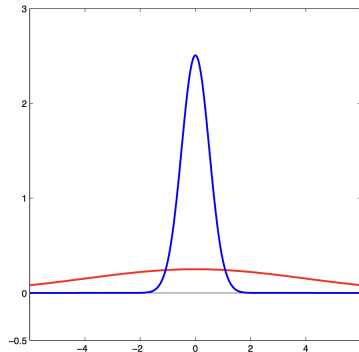
$\forall f \in L^2(\mathbb{R}),$

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with equality **if and only if** f is a constant multiple of the gaussian, e^{-kx^2} for $k > 0$.

What does the inequality tell us?



The Heisenberg-Weyl Inequality

Theorem: The Heisenberg Uncertainty Principle

Let ψ be a wave function such that

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1,$$

Let A be the position operator and let B be the momentum operator, such that $Af(x) = xf(x)$ and $Bf(x) = \frac{1}{2\pi i} f'(x)$.

The Heisenberg-Weyl Inequality tells us,

$$\text{Var}(A)\text{Var}(B) \geq C$$

for some fixed constant C .